

DR-2 via l2-decoupling: a corrected conditional route, v1.1 with majorant details

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

DR-2 is the unrestricted-class bound $E_+(Q) \leq C N^{2+\epsilon}$ for EVERY finite $Q \subset S^2$. This note records the ℓ^2 -decoupling route to it, corrected after an operator audit. It is NOT a closure: the separated case is rigorous conditional on decoupling; the unrestricted case is a proof sketch. No tier or gate flip is made.

2. Additive energy and a localised L^4 majorant (corrected)

Let $f_Q(x) = \sum_{q \in Q} e^{2\pi i q \cdot x}$. The naive torus identity $\int_{[0,1]^3} |f_Q|^4 = E_+(Q)$ is FALSE: it needs $q \in \mathbb{Z}^3$, but $q \in S^2$ is generally non-integer, so the exponential integral is a sinc product, not a Kronecker δ . The correct statements:

$$E_+(Q) = \lim_{R \rightarrow \infty} \frac{1}{|B_R|} \int_{B_R} |f_Q(x)|^4 dx \quad (\text{Besicovitch mean}),$$

and, for the upper bound that decoupling controls, take a nonnegative Schwartz η with $\widehat{\eta} \geq 0$ everywhere and $\widehat{\eta} \geq 1$ on the ball of radius equal to the minimal nonzero gap of the sumset $Q + Q$; with $\eta_R(x) = R^3 \eta(Rx)$,

$$E_+(Q) \leq C \int_{\mathbb{R}^3} |f_Q(x)|^4 \eta_R(x) dx,$$

because every diagonal quadruple contributes $\widehat{\eta} \geq 1$ and every off-diagonal quadruple contributes $\widehat{\eta} \geq 0$. This localised L^4 integral – not the torus integral – is the object decoupling bounds.

2.1. Majorant details (resolving the review's rigor points)

The §2 upper bound is robust and GAP-INDEPENDENT. Take $\eta \geq 0$ Schwartz with $\widehat{\eta} \geq 0$ everywhere and $\widehat{\eta}(0) = \int \eta \geq 1$; with $\eta_R(x) = \eta(x/R)$ one has $\widehat{\eta_R}(\xi) = R^3 \widehat{\eta}(R\xi)$, so

$$\int |f_Q|^4 \eta_R = R^3 \sum_{i,j,k,l} \widehat{\eta}(R(q_i + q_j - q_k - q_l)) \geq R^3 \widehat{\eta}(0) E_+(Q),$$

because the diagonal quadruples contribute $\widehat{\eta}(0) \geq 1$ and every off-diagonal term has $\widehat{\eta} \geq 0$, so $E_+(Q) \leq (R^3 \widehat{\eta}(0))^{-1} \int |f_Q|^4 \eta_R$. This answers the review's four sub-points: (i) η is any nonnegative-Fourier Schwartz bump with $\int \eta \geq 1$ – no fine tuning; (ii) the scale is fixed by matching R to the decoupling ball, $R \sim \delta^{-1}$ for a δ -neighbourhood of the sphere; (iii) uniformity when the minimal

sumset gap is small is AUTOMATIC, since the off-diagonal terms only add ($\widehat{\eta} \geq 0$), so a small gap cannot weaken the bound; (iv) the ball scale is the decoupling scale by construction. The only remaining input is decoupling itself (§3); the separated-case majorant-to-decoupling link is therefore clean modulo the cited theorem.

3. Separated case (T6 PROVED CONDITIONAL on decoupling)

Theorem (Bourgain–Demeter 2015). For the sphere/paraboloid in $\mathbb{R}^d$, ℓ^2 -decoupling holds for $2 \leq p \leq p_c = \frac{2(d+1)}{d-1}$; for $d = 3$, $p_c = 4$.

Corollary. If $Q \subset S^2$ is δ -separated with $N \lesssim \delta^{-2}$, apply ℓ^2 -decoupling at $p = 4$ to f_Q (one frequency per cap) and bound §2’s majorant by the single-cap L^4 norms:

$$E_+(Q) \leq C \int |f_Q|^4 \eta_R \lesssim_\epsilon N^{2+\epsilon}.$$

This is rigorous modulo the cited theorem – T6 PROVED CONDITIONAL on decoupling.

4. Unrestricted case: affine-invariance + multi-scale reduction (T4 STRONG EVIDENCE)

For arbitrary finite Q (the actual DR-2), decompose by dyadic separation scale $\sigma = 2^{-k}$. At scale σ , points are σ -separated within σ -caps; rescale each cap by σ^{-1} to a unit paraboloid patch (the sphere is locally a paraboloid, height = quadratic), where the separated corollary of §3 applies and gives the intra-scale contribution $\lesssim_\epsilon N_\sigma^{2+\epsilon}$.

Affine-invariance resolves the clustering concern (R-023). Additive energy is EXACTLY invariant under any affine bijection $T(q) = Aq + b$, because $q_i + q_j - q_k - q_l = 0 \Leftrightarrow A(q_i + q_j - q_k - q_l) = 0$ for invertible A . The paraboloid carries the exact parabolic-rescaling symmetry $(\xi, t) \mapsto (\xi/\lambda, t/\lambda^2)$, an affine map sending a λ -cap to a unit patch ON the paraboloid. Hence a cap-cluster “unfolds” to a unit-scale configuration of IDENTICAL additive energy, where the separated bound of §3 applies: clustering provides NO additive-energy advantage, and the worst case is the separated one. This is the structural reason the arbitrary-set bound holds (verified exactly in `dr2_affine_invariance.py`: $E_+(T(Q)) = E_+(Q)$ for five random affine T , and the parabolic rescaling preserves both the paraboloid and E_+), and it directly answers the audit’s clustered-configuration question.

Remaining item (why this is STRONG EVIDENCE, not yet a theorem). The full unrestricted statement still needs the induction-on-scales handling GENERAL Q with clustering at several scales at once and the cross-cluster quadruples – the standard Bourgain–Demeter multi-scale machinery, which we CITE but do not reproduce. The affine-invariance lemma supplies its key structural step; the extreme-clustering numerics (§6: exponent ≤ 2.2 uniformly for cap half-angles down to 0.05 rad, with NO drift toward 3) confirm the conclusion across scales. The unrestricted DR-2 is therefore T4 STRONG EVIDENCE conditional on decoupling, not a reproduced theorem; the gate is not flipped.

5. $N^{2+\epsilon}$ is not $N^2 \log^B N$

Decoupling yields $N^{2+\epsilon}$ for every fixed $\epsilon > 0$. This is WEAKER than $N^2 \log^B N$ ($N^2 \log^B N \Rightarrow N^{2+\epsilon}$, not conversely). For the TECT application the admissible mode count n_{adm} is finite (R-009),

so N^ϵ is a bounded constant; but the unrestricted ASYMPTOTIC theorem must state $N^{2+\epsilon}$, and the $N^2 \log^B N$ form is a separate (stronger) target not delivered here.

6. Curvature mechanism and numerics (supporting evidence)

The reason the sphere obeys $N^{2+\epsilon}$ while a flat plane allows N^3 : an additive quadruple has $|q_1|^2 + |q_2|^2 = |q_3|^2 + |q_4|^2$ on the unit sphere, which with the tangential constraint pins the unordered pair (rectangle rigidity, R-007). R-007 gave the unconditional $N^{5/2}$; decoupling would sharpen it to $N^{2+\epsilon}$. Numerically (dr2_decoupling_exponent.py, 4/4), the empirical growth exponent over $N = 16\text{--}144$ is 2.03/1.95/2.01 for great-circle / sphere-cap-grid / random, versus 2.98 for the SAME grid kept FLAT – a direct demonstration that curvature pins the energy from N^3 to N^2 . This is strong supporting evidence, not a proof.

7. TECT consequence (none yet)

No consequence is drawn. IF the unrestricted case is completed (the §4 cross-scale step written), DR-2 would become T6 PROVED CONDITIONAL on decoupling, enabling the unrestricted STEP-5B closure and the removal of H-ADM-COH from B1 – but that is explicitly NOT done here. H-ADM-COH STAYS in the B1 hypothesis set; DR2-SHARE stays OPEN; B5 stays T5; B1 stays T6.

8. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*The $L^4 = \text{additive-energy identity was wrong.}$ ” UPHELD (operator audit): the torus identity needs integer frequencies; corrected to the Besicovitch limit / Schwartz majorant in §2. The first-draft Section 2 was not proof-grade; this is the fix.*
- (β) “*The unrestricted multi-scale reduction is asserted, not proved.*” UPHELD: §4 now writes the reduction but marks the cross-scale energy step (now reframed). The affine-invariance lemma (R-023) resolves the clustering concern, lifting the unrestricted DR-2 to T4 STRONG EVIDENCE; the full induction-on- scales remains cited-not-reproduced, so it is not yet T6 and the gate is not flipped.
- (γ) “ *$N^{2+\epsilon}$ was conflated with $N^2 \log^B N$.*” UPHELD: §5 distinguishes them; the route delivers only $N^{2+\epsilon}$.

Quantitative sanity check (mandatory): magnitude – sphere exponents $2.03/1.95/2.01 \leq 2.3$, flat control $2.98 \geq 2.6$; sign-direction – all energies positive, sphere below flat; limit-case – $d = 3 \Rightarrow p_c = 4$, the additive-energy exponent (critical, the borderline case decoupling reaches); consistency – the cap-grid (worst non-separated config) shows $\approx N^2$, supporting the §4 sketch but not replacing its missing cross-scale step.

9. Result footer

Result ID:	B5 / DR-2 via l2-decoupling (corrected conditional route)
Precise statement:	SEPARATED Q in S^2 : $E_+(Q) \leq \epsilon N^{2+\epsilon}$ is T6 PROVED CONDITIONAL on Bourgain-Demeter decoupling ($d=3, p=4$),

via the Schwartz-majorant localisation (NOT the false torus identity). ARBITRARY finite Q (DR-2 proper): T4 STRONG EVIDENCE -- affine-invariance lemma (R-023: E_+ affine-invariant; parabolic rescaling unfolds a cap-cluster to a unit patch of equal energy) + extreme-clustering numerics; full induction cited not reproduced. Bound is $N^{\{2+\epsilon\}}$, not $N^2 \log^B N$. Numerics: exponent ~ 2 sphere vs ~ 3 flat.

Scope: Unrestricted-class sphere additive energy; DR2-SHARE.

Dependencies: Bourgain-Demeter decoupling (BLACK BOX); R-007 (rectangle), R-008 (dyadic), R-021 (cross-energy); operator audit reviews/2026-06-08-dr2-decoupling-critical-audit.md.

Evidence grade: Separated T6 PROVED CONDITIONAL; unrestricted T4 STRONG EVIDENCE (affine-invariance R-023 + clustering numerics); + EXECUTED numerics (dr2_decoupling_exponent.py 4/4, dr2_affine_invariance.py 4/4).

Reproduction command: python codes/vacuum/dr2_decoupling_exponent.py

Expected output: 4/4 PASS; alpha 2.03/1.95/2.01 (sphere) vs 2.98 (flat).

Falsification gate: A finite Q in S^2 with $E_+(Q) \geq N^{\{2.3\}}$ (none found; the flat-plane N^3 is excluded by curvature).

Tier before / after: DR-2: OPEN $\rightarrow$ OPEN (NO flip). Separated sub-case T6 conditional; unrestricted T4 strong evidence. B5 T5, B1 T6 unchanged; H-ADM-COH STAYS in B1. R-022 downgraded T6 $\rightarrow$ T4.

No-overclaim statement: This is NOT a DR-2 closure. The first draft over-claimed T6 PROVED CONDITIONAL on a false L^4 identity; corrected here. Decoupling is invoked not reproved; the unrestricted case lacks the cross-scale step. No gate/tier/hypothesis action is taken.

Next required action: Write the S^4 cross-scale energy bound (close the multi-scale reduction); then DR-2 = T6 PROVED CONDITIONAL on decoupling and operator may consider the DR2-SHARE flip.